Mehrabtronic Printer Control Center (PCC-4)

V1 Design Specification — Hardware & Software Architecture

1. Product Goal

A compact desktop control center for monitoring and controlling up to four network-connected Creality 3D printers from a single device.

2. Industrial / Mechanical Design

Parameter	Specification
Approx. dimensions	200 × 60 mm
Display	16×2 character LCD, centered
Control	Rotary encoder, positioned to the right of the LCD
Status panels	Two printer panels on the left + two on the right
Form factor	Linear, symmetric desktop design; intended to sit below a monitor

3. User Interface

The firmware is divided into two main UI layers: Status Layer for monitoring and Command Layer for commands and settings.

Status Layer — LCD Pages

Page 1 — Print Progress

```
P1:100% P2:63%
P3:25% P4:0%
```

Shows the print progress of all four printers.

Page 2 — Nozzle / Bed Temperature

```
1:215/60 2:220/65
3:0/0 4:205/55
```

Format: Nozzle / Bed temperature for all four printers.

Page 3 — Remaining Time

```
1:2h14 2:45m
3:-- 4:8h32
```

Shows remaining print time for all four printers.

Page 4 — Speed / Flow

```
1:100/95 2:80/100
3:--/-- 4:120/95
```

Format: Speed / Flow for all four printers.

Command Layer

A short encoder press enters the command menu: Select Printer → P1 / P2 / P3 / P4 → Pause / Resume / Stop / Preheat / Cooldown / Settings.

4. Smart Alert System

Important events such as print completion, faults, filament runout, or connection loss temporarily interrupt the current LCD page and display an alert. After acknowledgement, the UI returns to the previous page.

```
Examples: P3 — PRINT DONE | P2 — FILAMENT RUNOUT
```

5. Printer Status Indicators

Printer state	LED color
Printing	Green
Heating	Yellow
Paused	Amber
Finish	Blue
Fault	Red

Each printer has five dedicated status LEDs positioned beside the printed labels. Only the active state is illuminated at a time.

6. Encoder Behavior

Rotate: navigate pages/options   Short press: enter/select/confirm   Hold 2 s: return to Home

7. Network Architecture

The controller maintains four simultaneous WebSocket connections, one to each printer, providing real-time status updates.

```
Printer 1 ─┐
Printer 2 ─┼── ESP32
Printer 3 ─┤
Printer 4 ─┘
```

8. Printer IP Configuration

Settings → Network → Printer IP. The device requests the IP address of Printer 1, then Printer 2, Printer 3, and Printer 4 in sequence.

```
IP editing is performed by octet. Example: [192].168.1.10 → 192.[168].1.10 → 192.168.[1].10 → 192.168.1.[10] → Next.
```

Rules: each octet is limited to 0–255; the last saved IP is displayed when editing; first boot uses 192.168.0.0; the user can modify only the required octet.

After editing Printer 4, the final Next arrow returns the user to the previous menu.

9. Persistent Settings

Printer IP addresses are stored in the ESP32 NVS. After power-up, the controller connects to Wi-Fi, loads the saved addresses, and automatically reconnects all four WebSockets.

10. Hardware Architecture

Block	Selection / Role
MCU	ESP32-S3
Display	16×2 character LCD with I²C
Control	Rotary encoder with push button
Audio	Active buzzer
Power	USB-C 5 V input
Status LEDs	Standard diffused through-hole LEDs; no RGB / no WS2812
LED Driver	74HC595 shift registers

11. LED and PCB Architecture

Standard diffused through-hole LEDs are used because the LCD creates a mechanical height/gap between the main PCB and the front panel. The LEDs are placed beside the printed status labels rather than behind the labels.

The electronics are split into three boards:

Main PCB: ESP32-S3, 74HC595 shift registers, power circuitry, USB-C, buzzer, and board connectors.
Left LED PCB: status LEDs for P1 and P3.
Right LED PCB: status LEDs for P2 and P4.

The LED PCBs connect to the Main PCB through wired connectors. This reduces PCB cost, simplifies assembly, and makes replacement/repair easier.

74HC595 shift registers are used to expand GPIO outputs while keeping the ESP32 interface simple and scalable.

12. Firmware Architecture

The firmware is modular and object-oriented. Each module has a defined responsibility.

```
main.cpp → PrinterManager → Printer (×4) → UI / LCD / LED / Alert / Settings / Commands
```

Module	Responsibility
Printer	Independent printer object containing IP, status, progress, temperatures, time remaining, speed, flow,
PrinterManager	Maintains and coordinates the four simultaneous WebSocket connections.
UI	Controls UI state, pages, navigation, and menu logic.
LCD	Handles character LCD rendering.
LED	Controls the printer status LEDs.
Alert	Handles important events, alert priority/queue, and notification behavior.
Settings	Manages configuration and NVS persistence.
Commands	Builds and sends printer control commands.

13. Event-Driven Design Principle

Modules should remain as independent as practical. For example, a PRINT_FINISHED event is handled by the alert/event layer; the LCD, LEDs, and buzzer then apply their own responses. This keeps the firmware easier to extend and maintain.

14. V1 Design Principles

Simplicity: one rotary encoder controls the entire interface.
Cost efficiency: standard diffused LEDs, character LCD, and inexpensive 74HC595 drivers.
Manufacturability: modular, connectorized PCBs that are easier to assemble and service.
Scalability: hardware and software are structured so future features, additional printers, or support for other printer platforms can be added without redesigning the entire product.